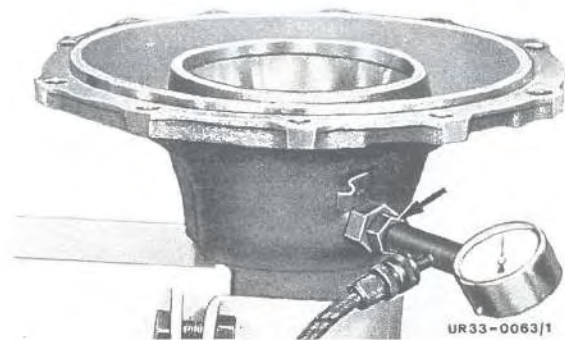
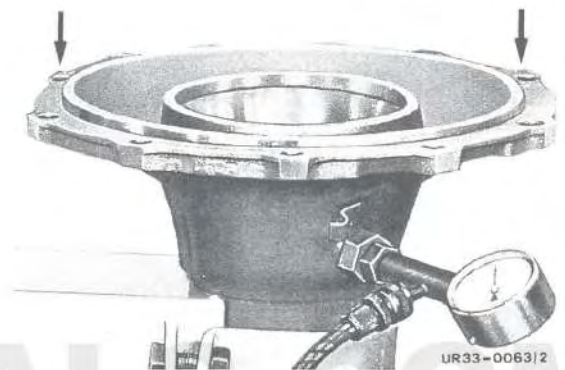


Checking

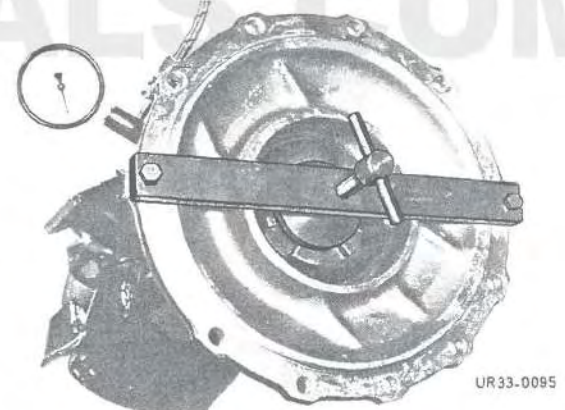
1 Connect pressure gage to differential lock.



2 Bolt special tool No. 22 to axle tube flange and apply pressure plate via spindle. Unscrew spindle by about 10 mm (this corresponds to position where differential lock is applied).



3 Connect air hose to pressure gage and admit 3 to 5 bar to piston several minutes after air supply is stopped.



Survey

Version

Rear axle 435.110/111/113/160/170

35.3

Rear axle 435.115/117

35.6

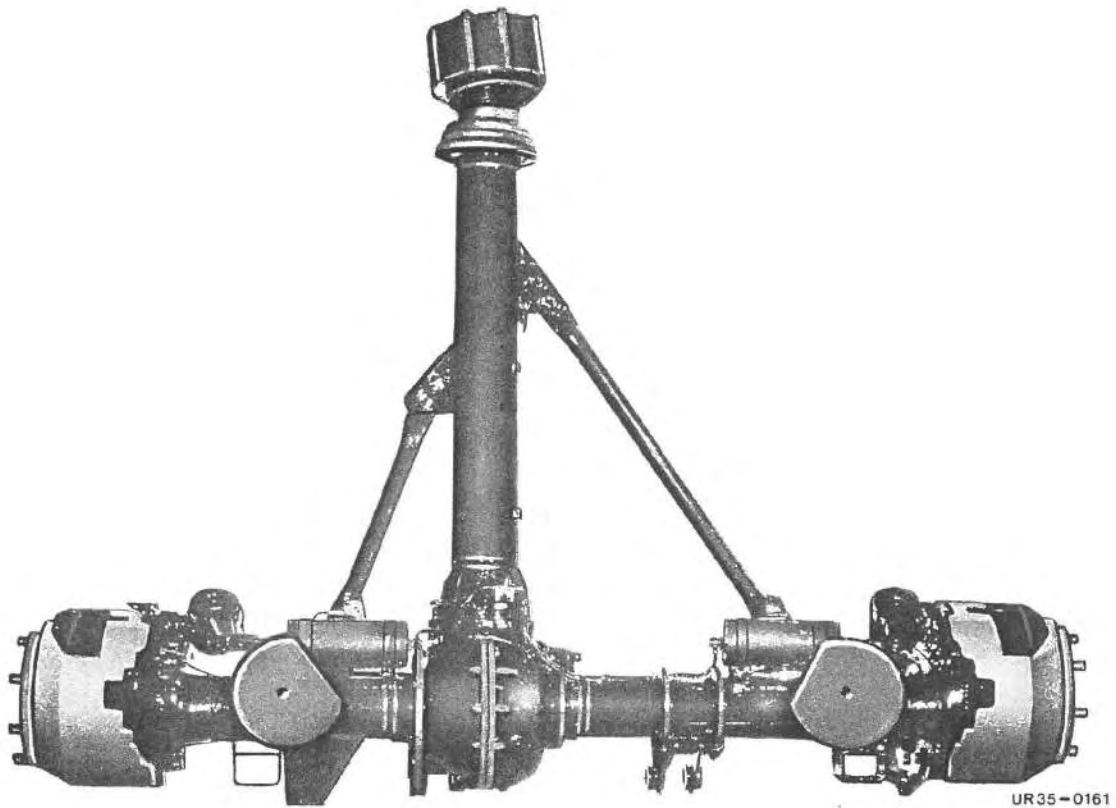
Contents

Chapter	Page
1 General	
Installation Survey	1.1/1
Technical Data	1.1/2
Ratios, Capacities	1.1/2
Key to Rear Axle Designation	1.1/2
Axle Sectional View	1.2/1
Setting Values, Necessary Materials	1.3/1
Special Tools	1.4/1
Tightening Torques	1.5/1
Exploded View	1.6/1
Worksheet for Adjusting Axle Drive	1.7/1
2 Removal and Installation of Rear Axle	2.1/1
3 Disassembly and Assembly of Rear Axle	3.1/1
4 Disassembly and Assembly of Drive Pinion Bearing	4.1/1
5 Disassembly and Assembly of Differential Gearing	5.1/1
6 Disassembly and Assembly of Wheel Hub Drive	6.1/1

Installation survey

Chassis		Rear axle		Brake	Installation
Model	Sales designation	Model	Sales designation		
435.110	U 1700	747.204	HU 3/1 S- 5,3	Disk brake	Series
.111	U 1700 L				
.160	Kit TM 170	747.231	HU 3/1 S- 6,8		
.170	Kit Condor	747.232	HU 3/1 S- 6,2		
435.111	U 1700 L	747.205	HU 3/1 S- 5,3		SA
435.113	U 1700 L/38	747.206	HU 3/1 S- 6,8		Series

Overall View



UR35-0161

Rear axle

35.3 General

747.2

Technical Data

Ratios

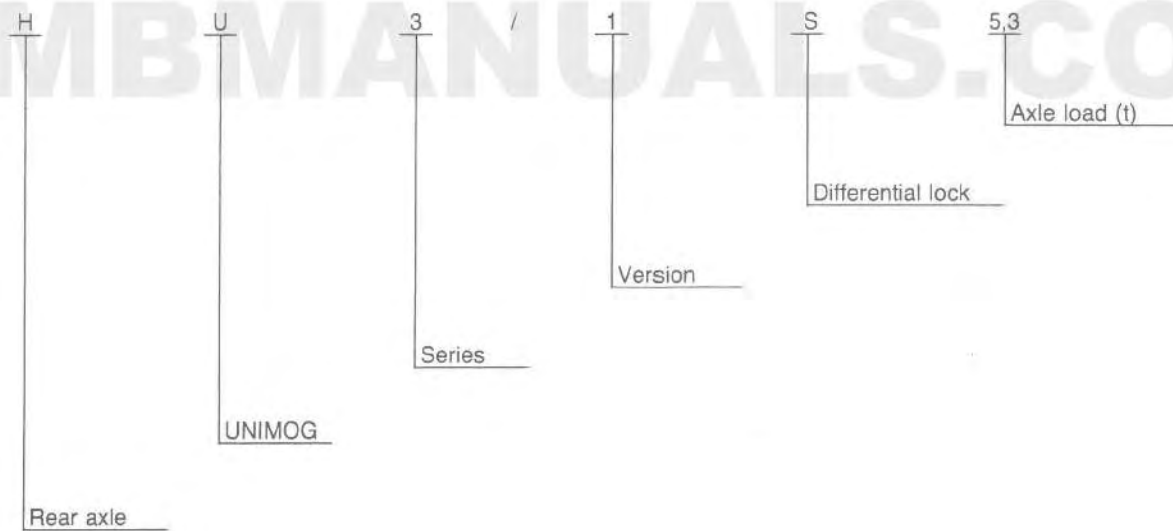
Axle model	Installation	Gear train Ring gear/drive pinion		Gear train Wheel hub drive		Total Ratio
		Number of teeth	Ratio	Number of teeth	Ratio	
747.204 .206 .230/.232 .231	Series	24: 11	2,18	38: 13	2,92	6,38
747.205	SA	24: 13	1,85			5,40
747.204	SA ¹⁾	25: 9	2,78			8,12

1) Only applies to model 435.110

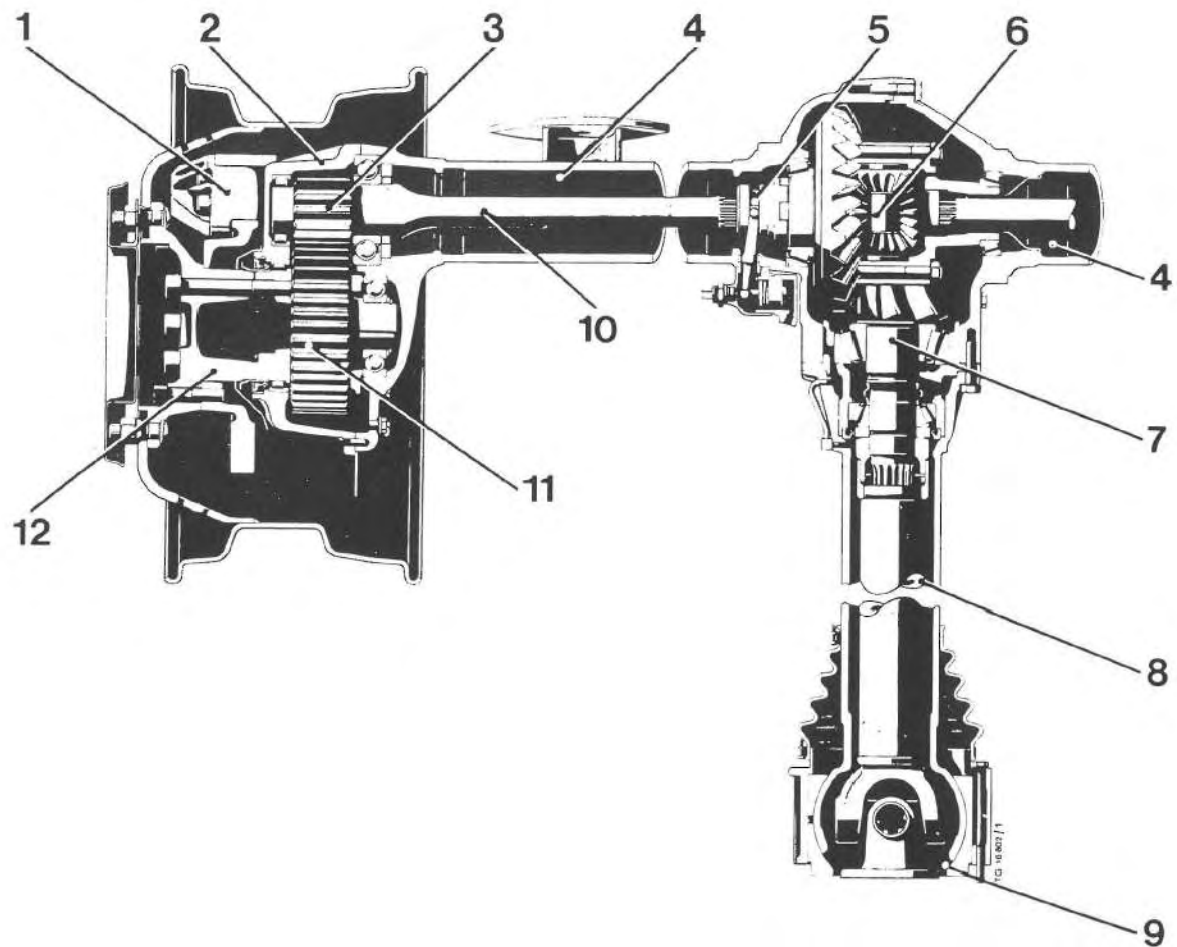
Capacities

Designation	Service product	SAE grade	Capacity (L)	Axle model
Axle center housing	Hypoid gear oil	90	2,5	747.204 .205 .230/.232 .231
Wheel hub drive			each 0,6	

Key to Rear Axle Designation



Sectional View



HU 3/1S-5.3

- 1 Fixed caliper
- 2 Wheel hub drive
- 3 Drive gear
- 4 Axle tube
- 5 Differential lock
- 6 Differential

- 7 Drive pinion
- 8 Torque tube
- 9 Torque ball casing
- 10 Axle shaft
- 11 Driven gear
- 12 Wheel hub

Setting Values

Designation	Setting Values
Preload of torque ball shells on extension tube mounting flange manual effort measured	200 to 300 N
Axle drive housing to large tapered roller bearing on lock end	± 0
Axle drive housing to small tapered roller bearing Preload (refer to axle data sheet)	0,2 mm
Axle tube to pinion flange	± 0
Deep-groove ball bearing to driven gear in wheel hub drive	± 0
Breakaway force for drive pinion, (refer to axle data sheet)	6,0 to 6,5 nm

Necessary Materials

Current No.	Designation	Part Number
1	Teroson Atmosit sealing compound	000 989 43 20 ¹⁾
2	Longlife grease	commercially available
3	Molybdenum disulphide running-in paste	commercially available
4	Loctite No. 241 adhesive	000 989 70 71
5	Loctite No. 270 adhesive	002 989 93 71
6	Molykote liquid grease	commercially available
7	Molykote paste D X	commercially available
8	Hypoid gear oil SAE 90	commercially available
9	Terostat 56 surface sealing compound	001 989 58 20

1) optional 001 989 58 20

General note: In order to ensure satisfactory adhesion of sealing compounds, the sealing surfaces must be completely free of oil and grease.

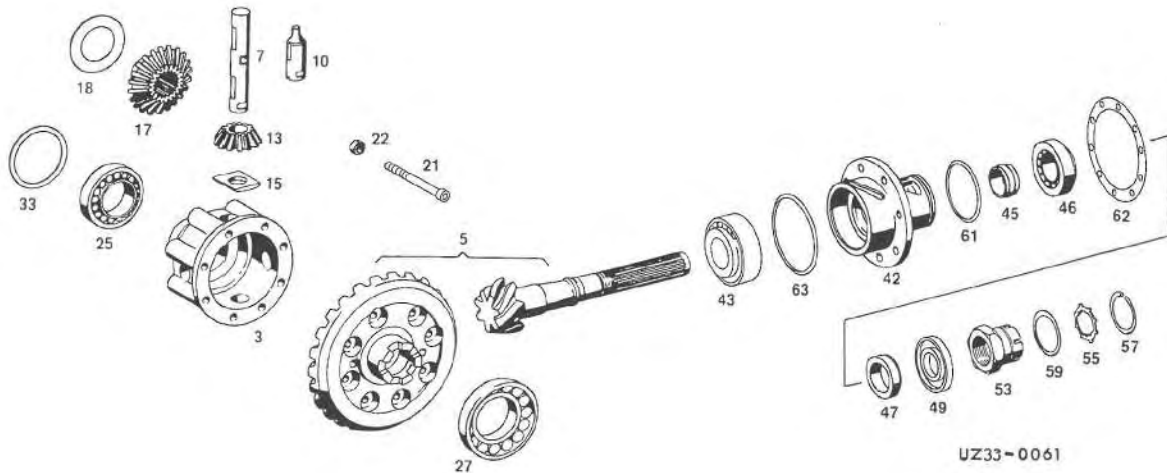
Special Tools

Cons. No.	Tool	Number	Tool set
1	Extension tube for mounting flange	000 589 01 14 00	A
2	Allan key socket 14 mm, 3/4" square for brake disk	001 589 68 09 00	B
3	Allan key socket 17 mm, 3/4" square for wheel hub	001 589 69 09 00	B
4	Remover and installer for driven gear	425 589 04 14 00	B
5	Guide pin (2 required) for wheel hub	425 589 01 15 00	B
6	Guide pin (3 required) for wheel hub drive housing	425 589 02 15 00	B
7	Drift for installer	406 589 04 15 00	B
8	Installer (4 required) for driven gear	425 589 02 33 00	B
9	Counter-support for pullers and extractors	000 589 35 33 00	B
10	Extractor for brake pads	001 589 43 33 00 (115 589 14 33 00)	B
11	Puller for counter-support (with tool No. 11)	425 589 00 34 00	B
12	Installer for deep-groove ball bearing and shaft seal in wheel hub drive (with tool No. 7)	425 589 00 43 00	B
13	Assembly flange for torque ball casing (with tool No. 1)	425 589 00 59 00	B
14	Torque wrench for drive pinion (4 to 16 Nm, 1/4" square)	000 589 67 21 00	C
15	Adjuster for drive pinion (with tool No. 18)	425 589 00 21 00 (406 589 10 21 00)	C
16	Mounting for drive pinion	425 589 00 31 00 (406 589 01 31 00)	C
17	Assembly sleeve for taper roller bearing in drive pinion	425 589 00 35 00	C
18	Assembly sleeve for swivel joint	425 589 01 35 00	C
19	Installer for roller bearing in wheel hub drive housing (with tool No. 18)	425 589 01 43 00	C
20	Installer for taper roller bearing in axle tube (with tool No. 7)	425 589 02 43 00	C
21	Installer for shaft seal in drive pinion (with tool No. 21)	425 589 04 43 00	C

Tightening torques

Location	Thread	Nm
Wheel hub to drive gear	M 20 x 1,5	600
Drive gear to axle shaft	M 20 x 1,5	600
Fixed caliper mounting to axle	M 20 x 1,5	500
Wheel bolt	M 20 x 1,5	400
Shock absorber mounting to axle	M 20 x 1,5	400
Brake disk to wheel hub	M 18 x 1,5	400
Axle strut to axle tube	M 18 x 1,5	350
Axle strut to torque tube	M 18 x 1,5	350
Rear spring to spring bracket	M 18 x 1,5	300
Transverse link to frame	M 16 x 1,5	300
Differential assembly	M 16 x 1,5	300
Axle tube assembly	M 14 x 1,5	200
Wheel hub drive housing to axle tube	M 14 x 1,5	200
Torque tube to pinion flange	M 14 x 1,5	200
Propeller shaft to transmission	M 12 x 1	100
Torque ball casing to transmission	M 10 x 1	60
Bearing ring to axle tube	M 10	40

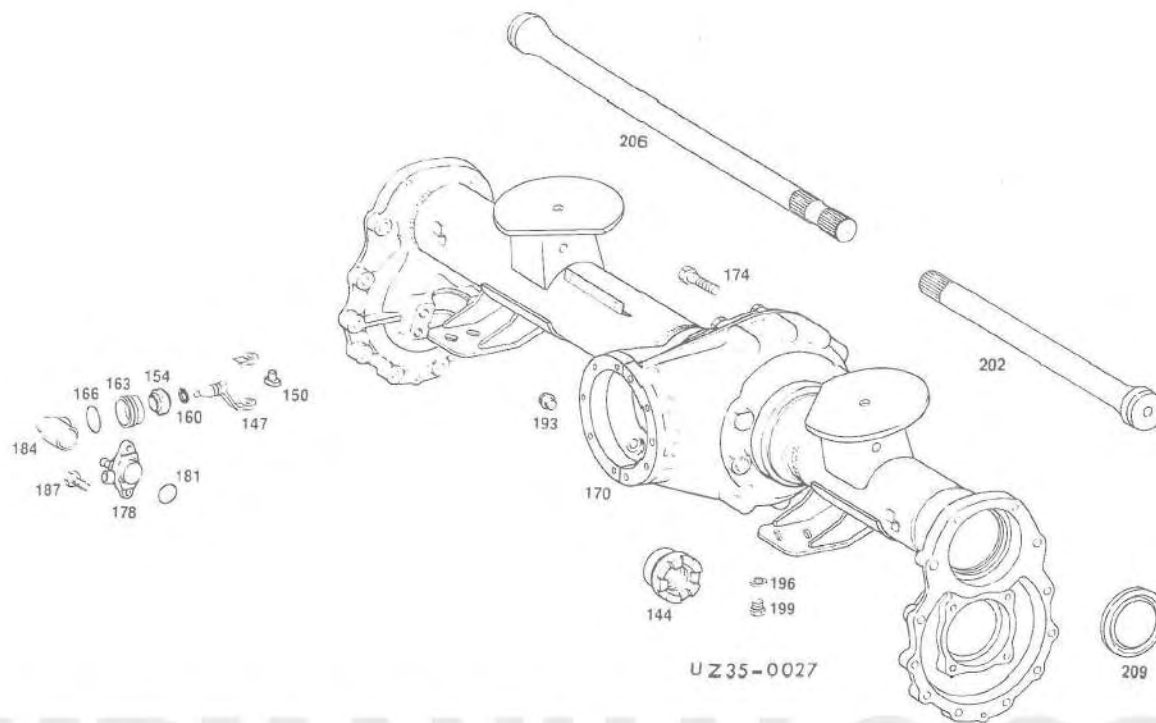
Exploded View



Differential housing and Drive Pinion

- | | |
|------------------------------|-------------------------|
| 3 Housing | 42 Pinion flange |
| 5 Gear set | 43 Taper roller bearing |
| 7 Differential pinion shaft | 45 Spacing sleeve |
| 10 Differential pinion shaft | 46 Taper roller bearing |
| 13 Bevel gear | 47 Wear ring |
| 15 Thrust washer | 49 Sealing ring |
| 17 Side gear | 53 Nut |
| 18 Thrust washer | 55 Retainer |
| 21 Bolt | 57 Snap ring |
| 22 Nut | 59 Sealing ring |
| 25 Taper roller bearing | 61 Sealing ring |
| 27 Taper roller bearing | 106 Shim |
| 33 Shim | 108 Sealing ring |

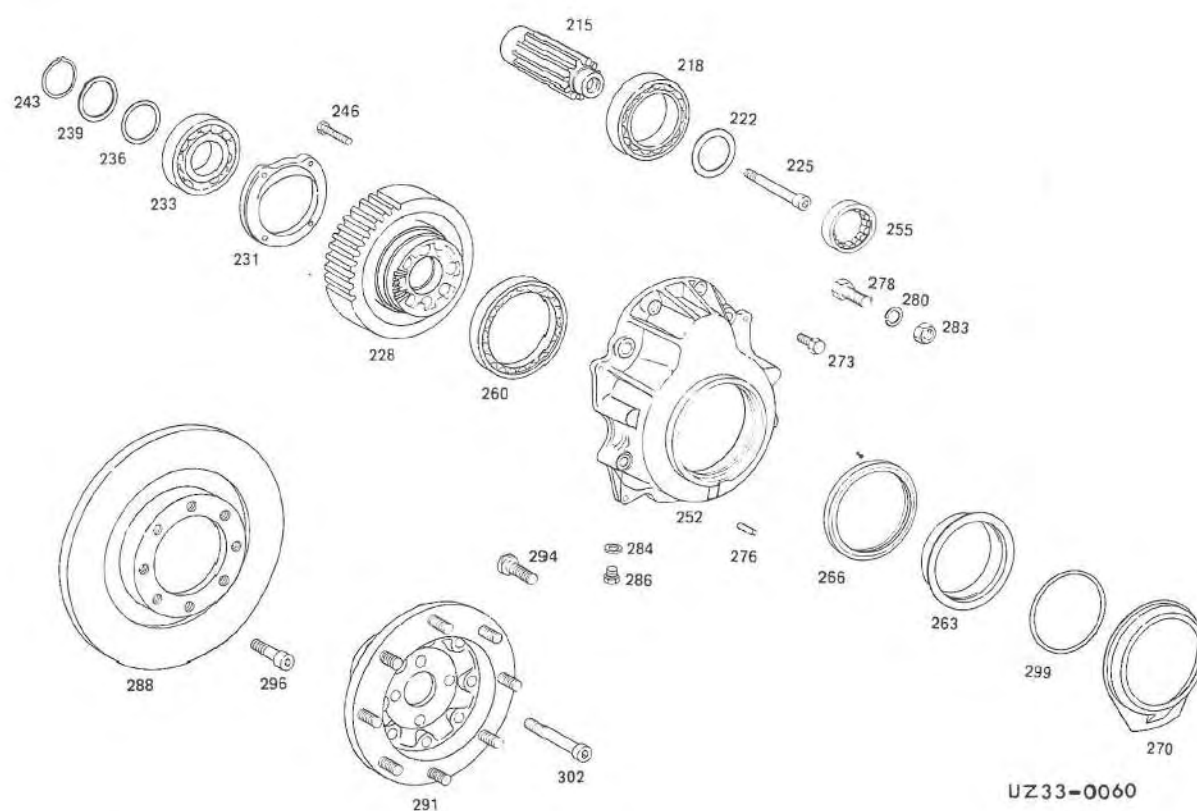
Exploded View



Axle tube and torque tube flange

- | | | | |
|-----|----------------|-----|------------------------|
| 144 | Shift dog | 181 | Sealing ring |
| 147 | Selector fork | 184 | Housing |
| 150 | Sliding block | 187 | Bolt |
| 154 | Swivel joint | 193 | Screw plug |
| 160 | Snap ring | 196 | Sealing ring |
| 163 | Bushing | 199 | Screw plug |
| 166 | Sealing ring | 202 | Rear axle shaft, left |
| 170 | Axle tubes | 206 | Rear axle shaft, right |
| 174 | Bolt | 209 | Spacing ring |
| 178 | Shift cylinder | | |

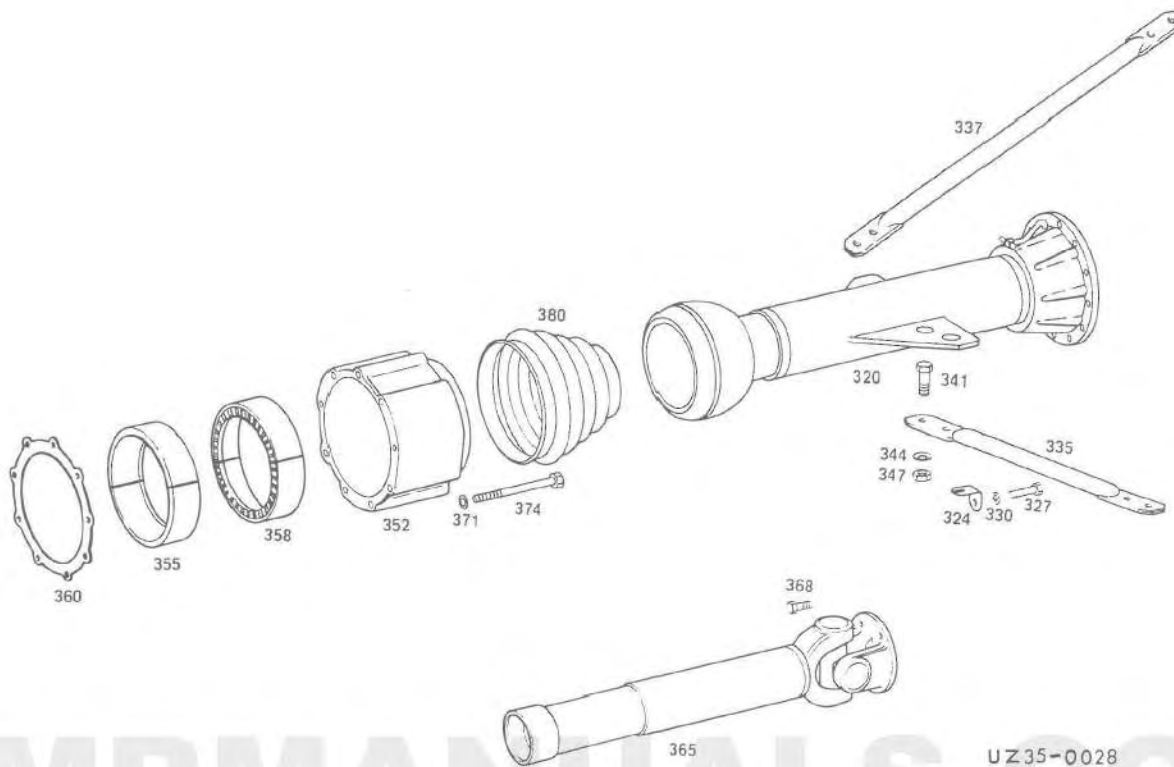
Exploded View



Wheel hub drive

215	Drive gear	266	Sealing ring
218	Ball bearing	270	Oil baffle
222	Shim	273	Bolt
225	Bolt	276	Straight pin
228	Driven gear	278	Bolt
231	Bearing ring	280	Lock washer
233	Ball bearing	283	Nut
236	Shim	284	Sealing ring
239	Shim	286	Screw plug
243	Snap ring	288	Brake disk
246	Bolt	291	Wheel hub
252	Housing	294	Wheel stud
255	Cylindrical roller bearing	296	Bolt
260	Cylindrical roller bearing	299	Sealing ring
263	Wear ring	302	Bolt

Exploded View

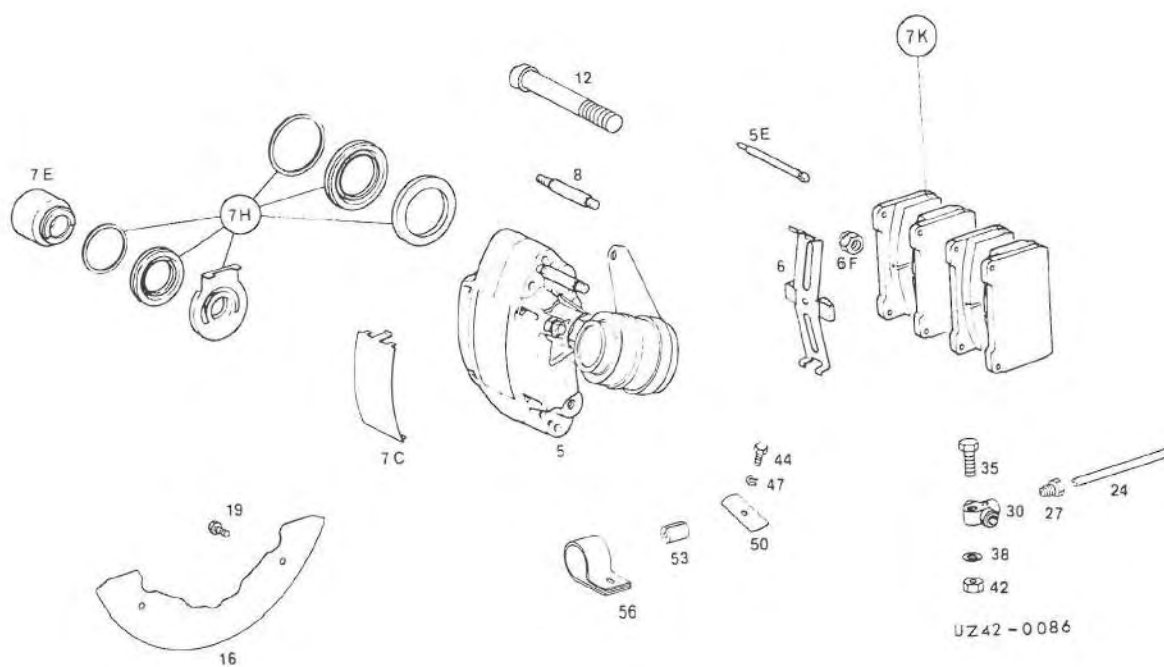


UZ35-0028

Thrust tube

- | | |
|---------------------------|----------------------------|
| 320 Thrust tube | 352 Torque ball casing |
| 324 Bracket | 355 Torque ball shell |
| 327 Bolt | 358 Torque ball shell half |
| 330 Spring washer | 360 Shim |
| 335 Thrust arm, left | 365 Propeller shaft |
| 337 Thrust arm, right | 368 Bolt |
| 341 Bolt | 371 Sealing ring |
| 344 Conical spring washer | 374 Bolt |
| 347 Nut | 380 Rubber gaiter |

Exploded View



Brake components

5	Fixed caliper
5E	Pin
6	Spring clip
6F	Nut
7C	Metal cover
7E	Piston
7H	Sealing set
7K	Repair set, brake pads
8	Bleeder valve
12	Fixed caliper fastening bolt
16	Backplate
19	Bolt

24	Line
27	Nut
30	Junction
35	Bolt
38	Lock washer
42	Nut
44	Bolt
47	Lock washer
50	Shim
53	Hose
56	Clip

Worksheet for Adjusting Axle Drive

Axle model 737.2 – 747.2
737.3 – 747.3

UNIMOG
MB - trac

Customer

Order No.

Date

Dealer

first registered

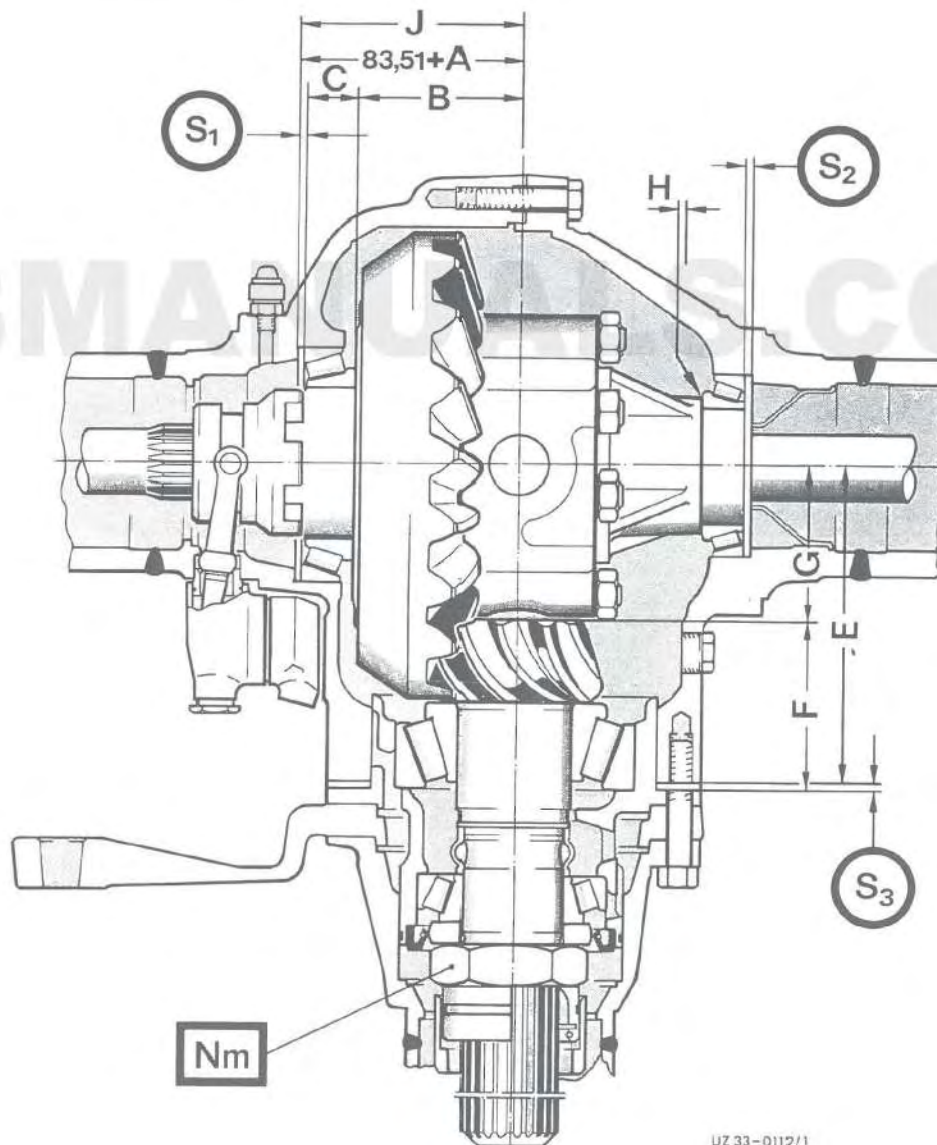
Chassis No.

Axle No.

Operating hours/odometer reading

Last service

carried out by

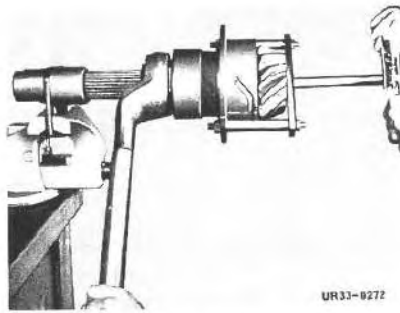


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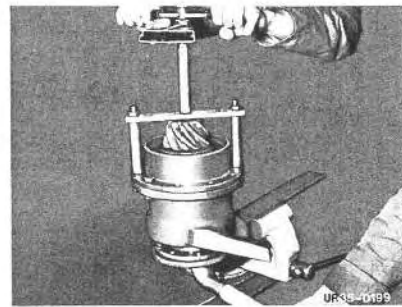
35.3 Adjusting Axle Drive

Nm

Adjust breakaway force
for drive pinion bearing



Front axle

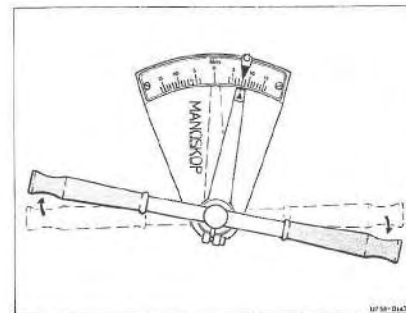


Rear axle

Measure and adjust preload

- Use new bearings
- Remove preservative from bearing and oil
- Only use new clamping sleeve
- Exactly maintain breakaway force!

Axle model	Breakaway forces
737.2/747.2	6,0 to 6,5 nm
737.30/747.30	7,5 to 8,0 nm
737.31/747.31	6,5 to 7,0 nm



Use torque wrench

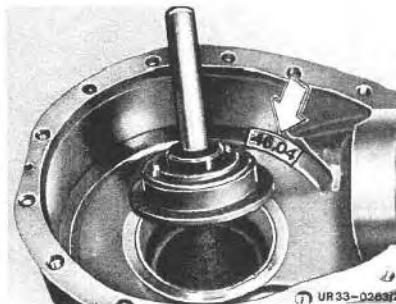
S₁

Calculate shim thickness S₁ (large bearing)

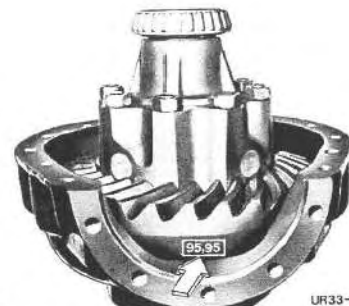
Enter dimensions here

Example

Dimension (const.) =	83,51	(83,51)
+ Dimension A =		(46,04)
Dimension J = mm		(129,55)



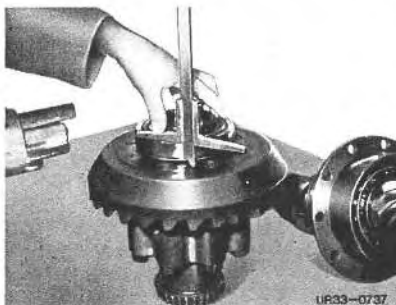
Read off dimension A at housing



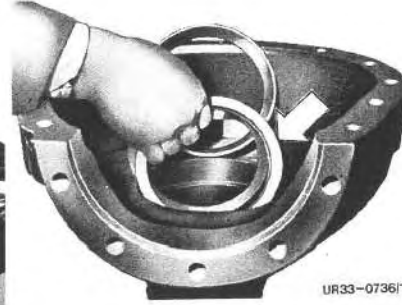
Read off dimension B at ring gear

Dimension A =		(95,95)
+ Dimension C =		(32,00)
Dimension B+C = mm		(127,95)

Dimension J =		(129,55)
- Dimension B+C =		(127,95)
Shim S ₁ = mm		(1,60)



Measure dimension C (bearing height)



Insert shims S₁ mm

S₂

Calculate shim thickness S_2 (small bearing)

- Install outer bearing race **without** shims.
- Mount bearing on differential housing, press in to approx. 4 mm from bearing seat
- Install differential gearing in axle housing.
- Assemble both axle tubes without drive pinion, fasten with some screws (200 Nm) and release once again.
- Measure gap H using feeler gauge, add 0.2 mm preload dimension I .
- Once again, withdraw outer bearing race and insert shims S_2 .

Enter dimensions here

Example

	Dimension H =	(1,40)
+ Preload	Dimension I = 0,20	(0,20)
Shim = mm		<u>(1,60)</u>



UR33-0735/1

Measure dimension H

S₃

Calculate shim thickness S_3 (flange-drive pinion bearing)

Enter dimensions here

Example

	Dimension F =	(101,60)
+ Dimension	G =	(93,20)
Dimension $F+G$ = mm		<u>(194,80)</u>



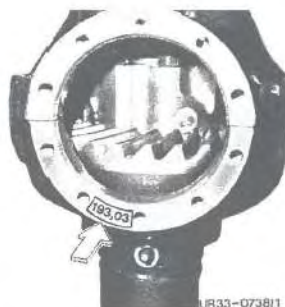
UR33-0728

Measure dimension F



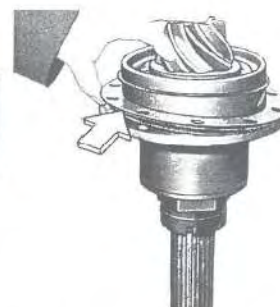
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Read off dimension G



UR33-0738/1

Read off dimension E at housing



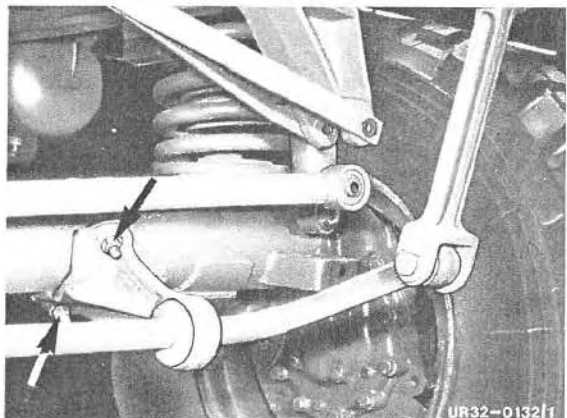
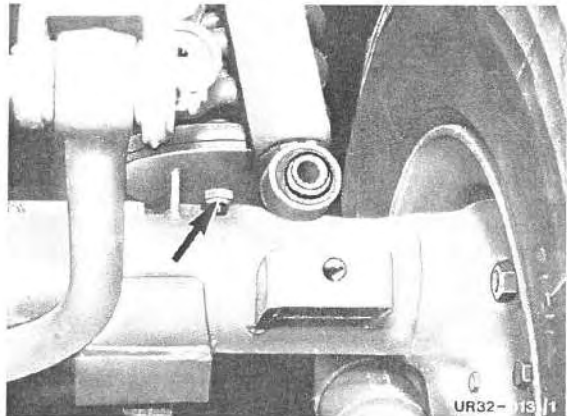
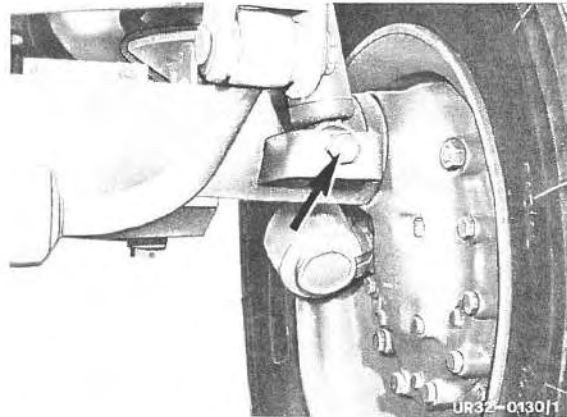
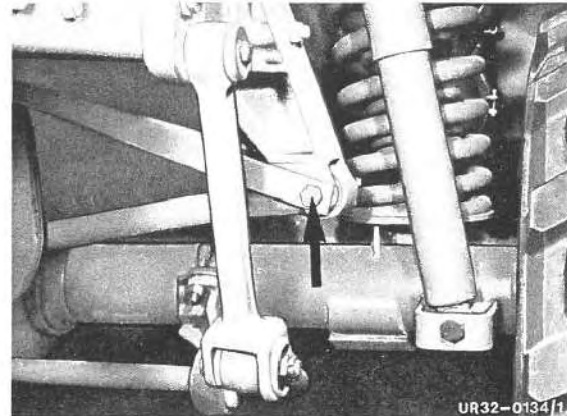
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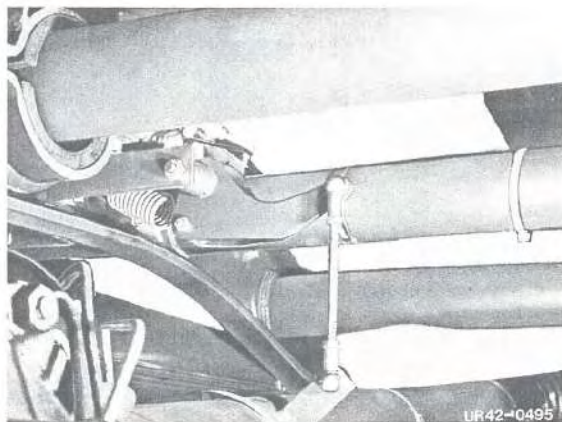
Insert shims S_3 mm.

	Dimension $F+G$ =	(194,80)
- Dimension	E =	(193,03)
Shim S_3 = mm		<u>(1,77)</u>

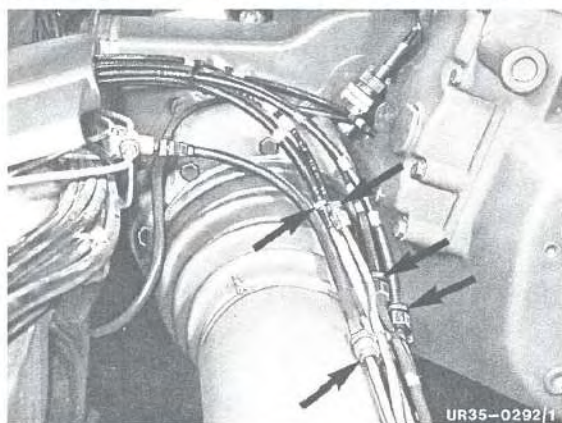
Removal

- 1 Remove spare wheel.
- 2 Unscrew transverse link.
- 3 Unscrew shock absorber at axle.
- 4 Unscrew both springs.
- 5 Unscrew stabilizer at both brackets.

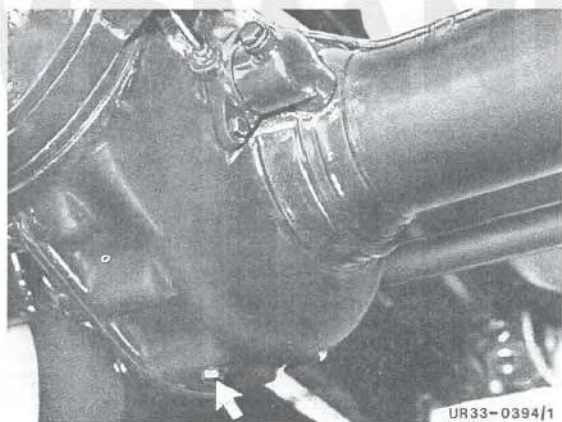




- 6 Release and unhook pull rod of ALB modulator.



- 7 Release line holder. Detach brake line, air supply and exhausting lines, as well as line to spring-loaded cylinders.



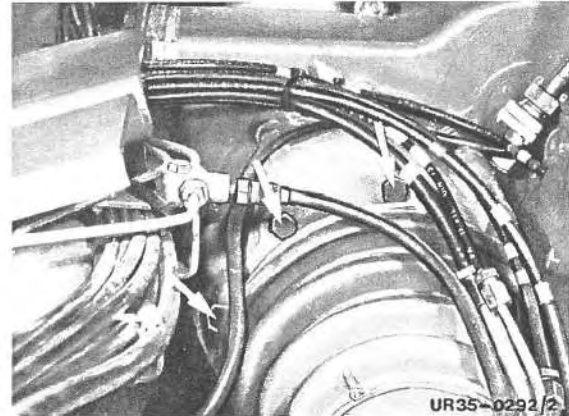
- 8 Drain oil off center axle housing.



- 9 Drain oil off wheel hub drive.

10 Release rubber gaiter and unscrew torque ball casing.

11 Raise vehicle, release springloaded cylinder, draw axle to rear and support torque tube.

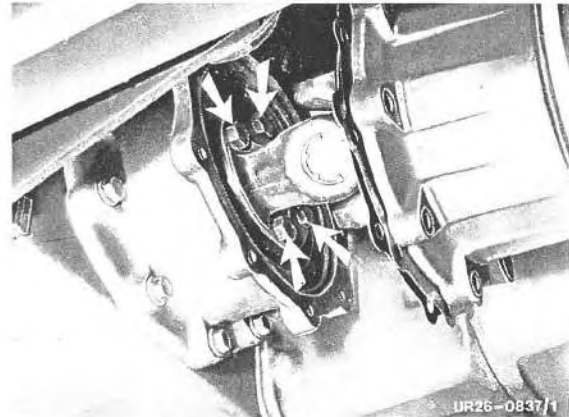


12 Remove front torque ball shell halves.

13 Unscrew propeller shaft at transmission.

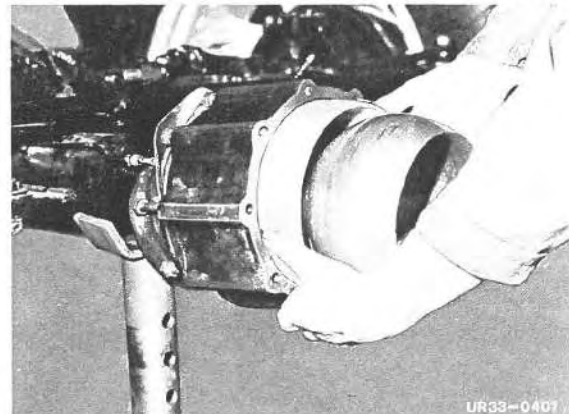
14 Lift vehicle off springs and draw axle out to rear.

15 Deposit vehicle on trestles.



16 Draw torque ball casing back and remove rear torque ball shell halves.

17 Clean and check torque ball shell halves and torque ball casing, exchange if necessary.



Preparation for installation

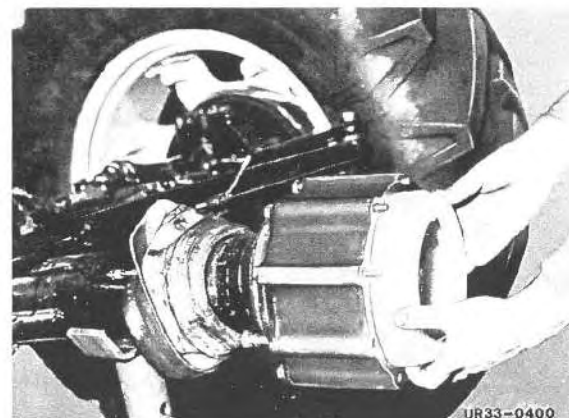
1 **Carefully** deburr torque ball casing and torque ball shell halves.

2 Coat rear torque ball shell halves with paste No. 3 prior to installation.

3 Coat front torque ball shell halves with paste No. 3 prior to installation.

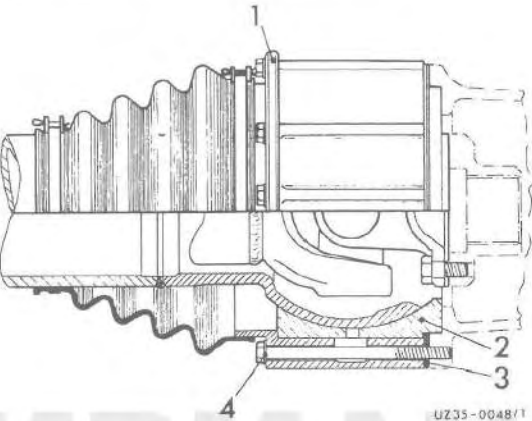
Note:

From Model	Chassis end No.	Rear axle end No.
435.110		
.111	076 650	000 533
.113		
.170		



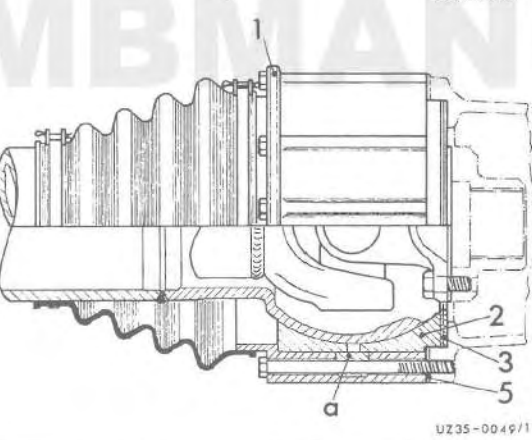
The shims, the torque ball housing and torque ball bearing shell were modified as follows:

Item	Designation	Modification
1	Torque ball housing	Recess in housing eliminated (see ill., Item a)
2	Torque ball bearing shell front	Overall length shortened from 66 mm to 62 mm
3	Shims	Change in shape, see ill.
4	Sealing ring A 10 x 16	For screw head/torque ball housing eliminated
5	Sealing compound Terostat 56	Used as seal between torque ball housing and trans- mission flange

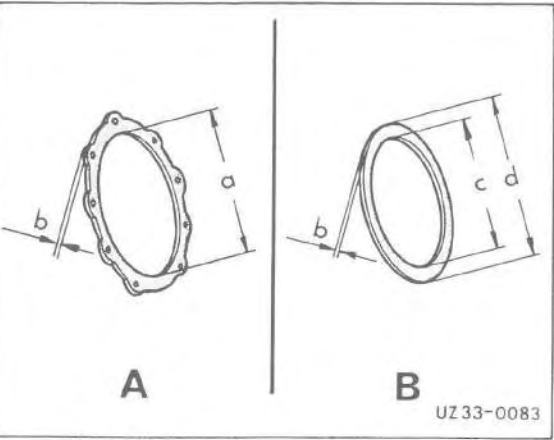


4 Position of shims (3) previously

Note:
When installing SA 35 973 and SA 35 974 (front and rear axle parts with fording ability), the sealing ring previously installed is not required.



5 Position of shims (3), new



6 Install shims for measuring.

- Note:**
- A Shims, previous
 - B Shims, new
 - a Dia. 205 mm
 - b 0.2; 0.5; 1.0 mm
 - c Dia. 148 mm
 - d Dia. 194 mm

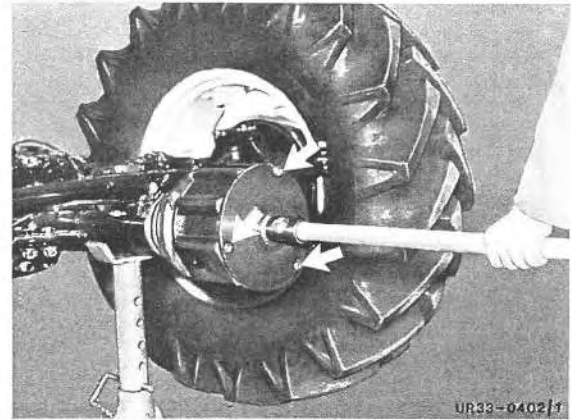
4 Attach special tool No. 13 and evenly tighten fastening bolts until a manual force of 200 to 300 N is reached at the extension tube.

Note: Remove shims if manual force required is inadequate, and add if excessive.

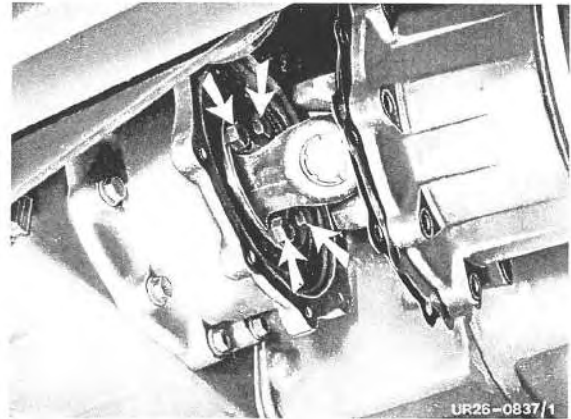
If using the new shims, shims must be inserted if the manual force required is inadequate. If the manual force required is excessive, remove shims.

Installation

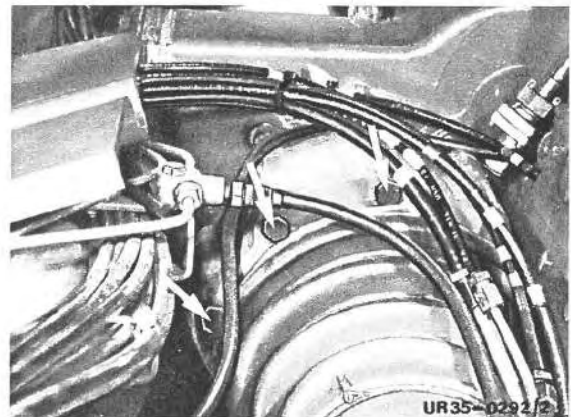
- 1 Raise vehicle and move axle into position.
- 2 Attach propeller shaft.
- 3 Attach and tighten torque ball housing with shims of calculated thickness.
See 1.5/1 for tightening torques.
- 4 Fit brake line, vent and bleeder line as well as line to spring loaded cylinders and screw on bracket.



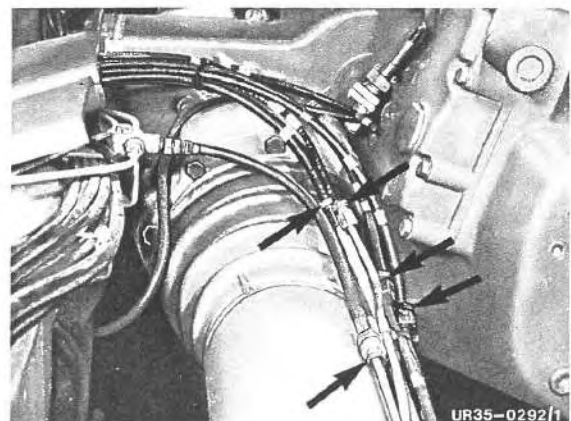
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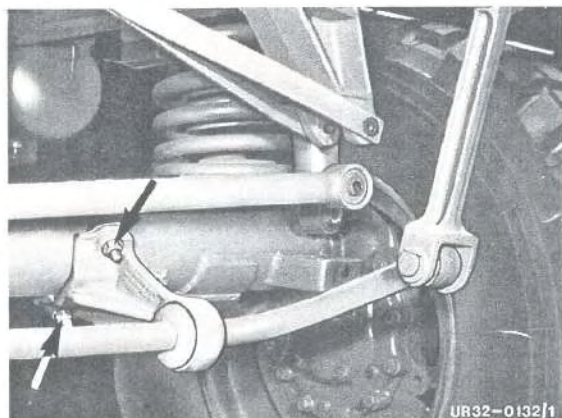
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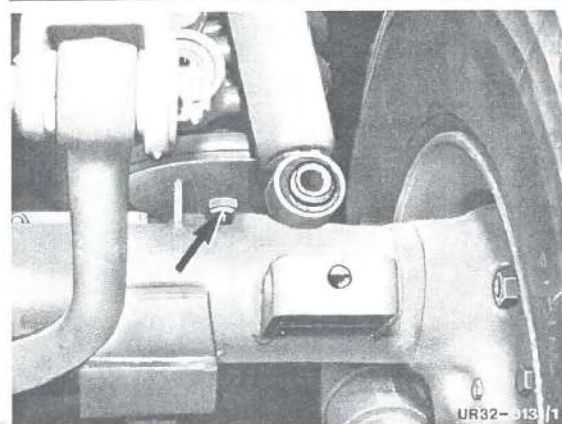
UR35-0292/2



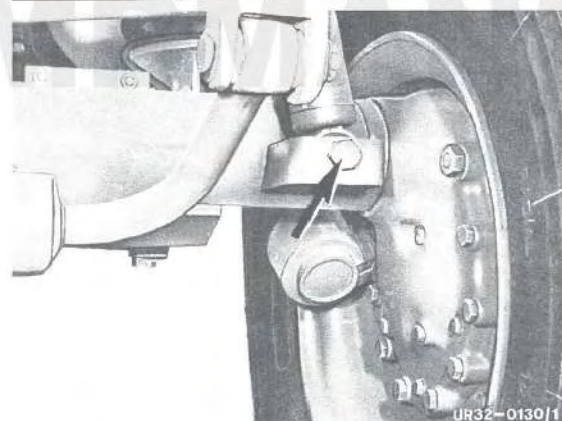
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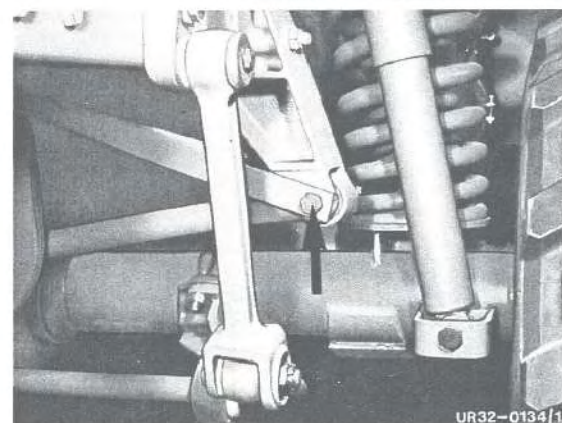
5 Screw on stabilizer.



6 Screw on both springs at bottom.

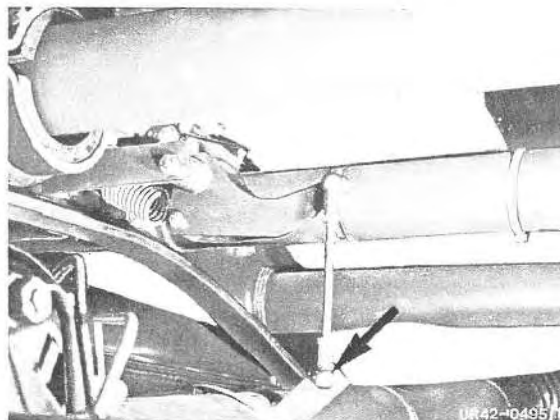


7 Screw on shock absorber.

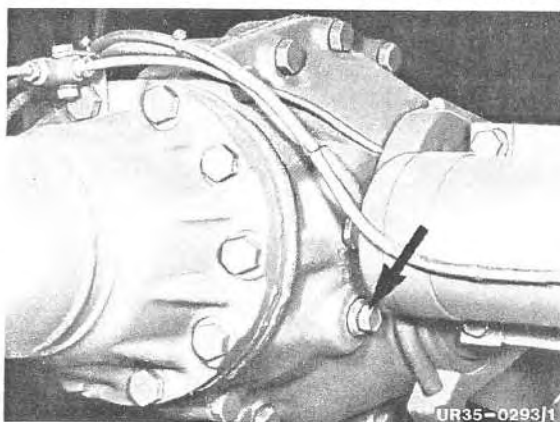


8 Screw transverse link onto frame.
See 1.5/1 for tightening torques.

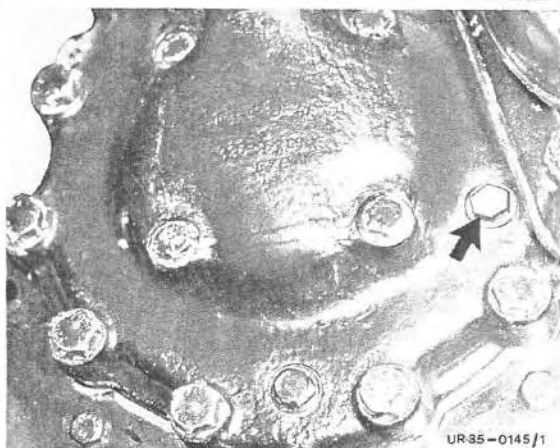
- 9 Attach tie rod of ALB regulator and secure.



- 10 Fill oil into axle centre housing.
See 1.1/2 for capacity.



- 11 Fill oil into wheel hub drive.
See 1.1/1 for capacity.
- 12 Bleed brake system.
- 13 Check brake system for function and leaks.



Disassembly

1 Clean rear axle, clamp in trestle and detach wheels.

2 Unscrew brake lines, axle ventilation line and differential lock line.

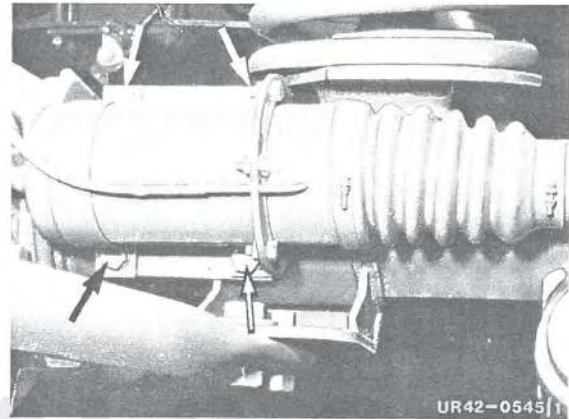
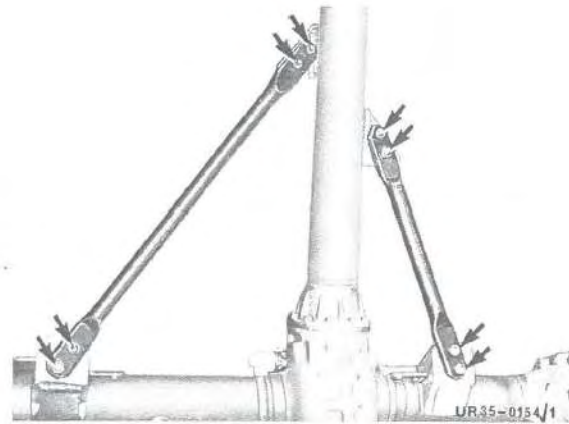
Note:

From Model Chassis end No. Rear axle No.
435.111 001 178

Axle ventilation is installed in axle tube on right-hand side.

3 Unscrew axle struts.

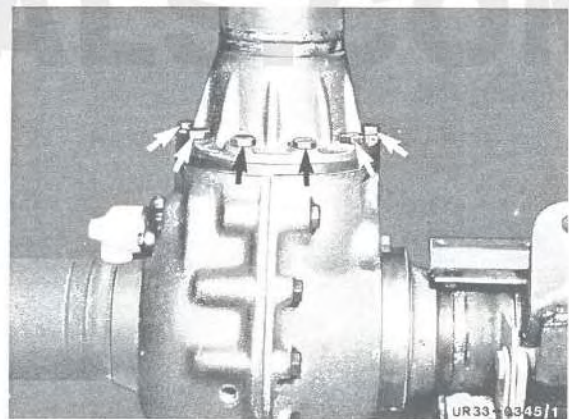
4 Detach spring-loaded cylinder.



5 Take propeller shaft out of torque tube, unscrewing and removing torque tube.

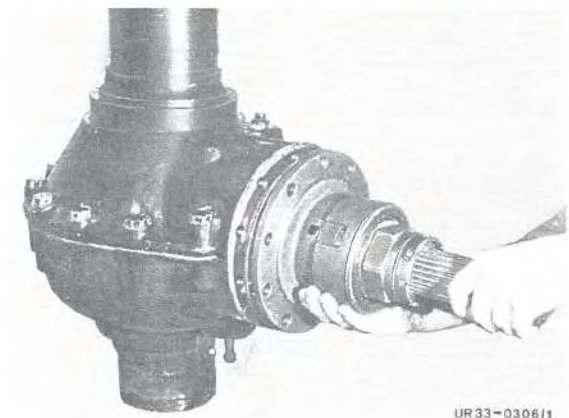
6 Disassemble wheel hub drive, referring to 6.1/1.

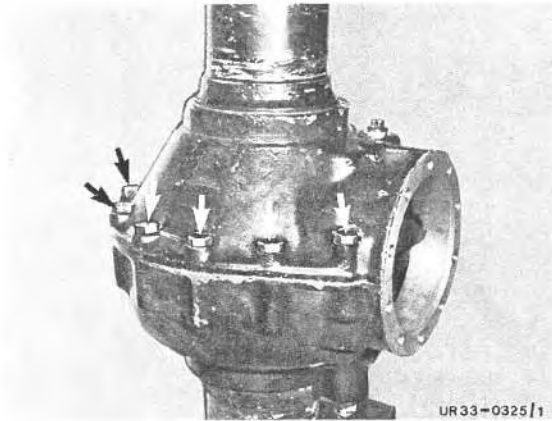
7 Turn axle by 90° in trestle.



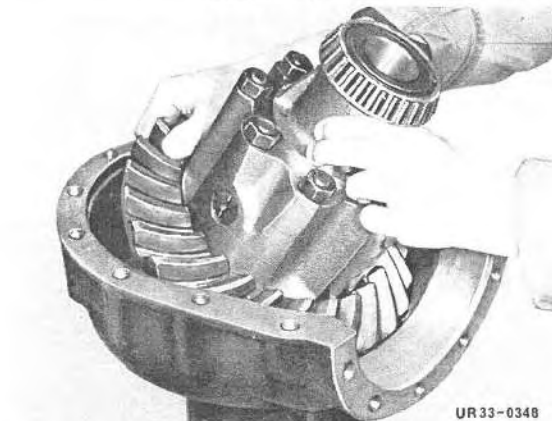
8 Release connecting bolts of both axle tubes and remove pinion flange with drive gear.

Note: Watch out for shims between pinion flange and axle housing.

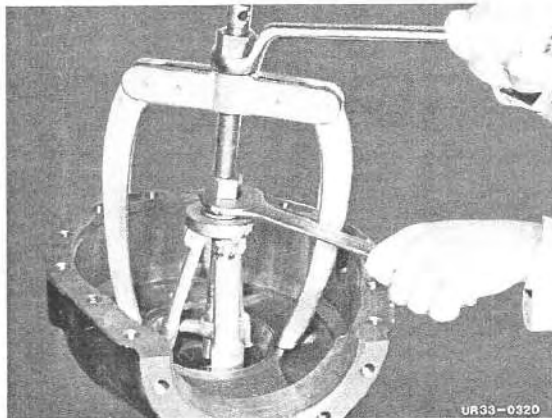




9 Disconnect axle tube at axle drive.

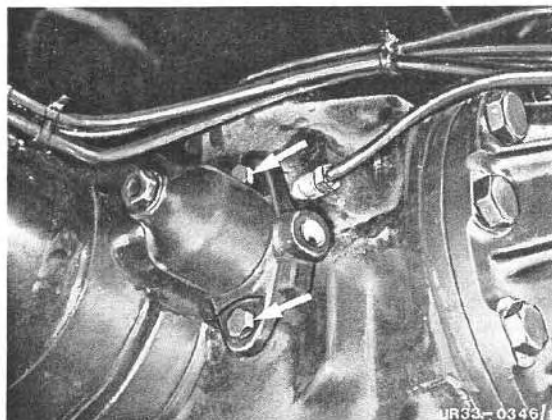


10 Take differential out of axle tube.



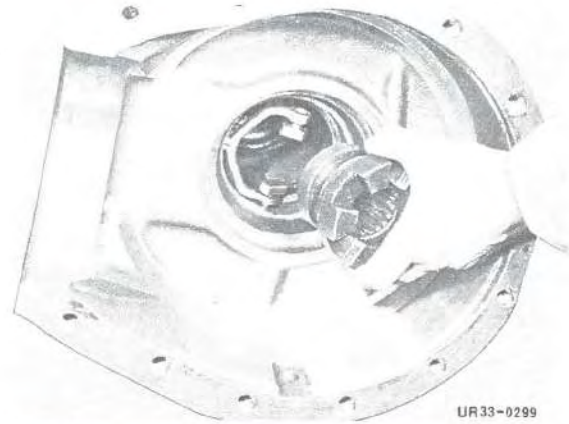
11 Using puller and special tool No. 9, pull outer taper roller bearing races out of both axle tubes.

12 Turn axle tube by 90° in trestle.



13 Unscrew and remove differential lock shift cylinder.

- 14 Take shift dog out of axle tube.

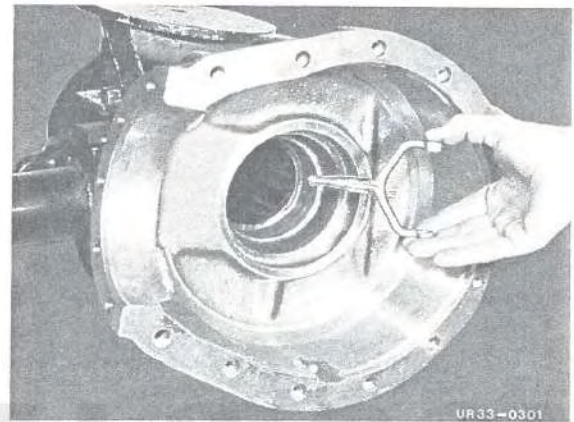


UR 33-0299

- 15 Remove snap ring from selector fork and take selector fork out of swivel joint.

- 16 Knock swivel joint out.

- 17 Clean and check all parts, exchange if necessary.

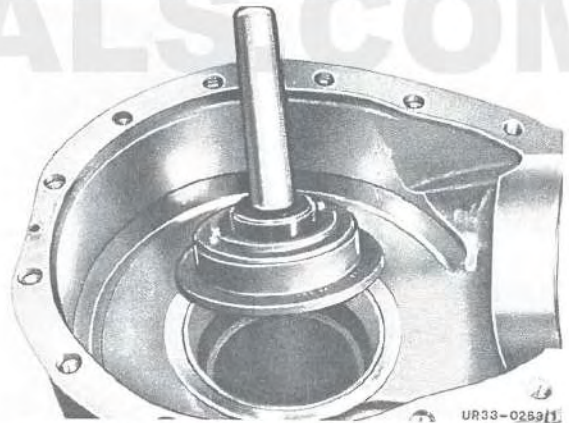


UR 33-0301

Assembly

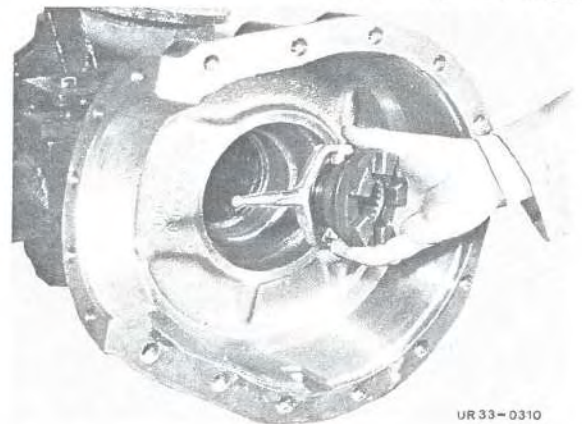
Note: Prior to assembling axle, adjust axle drive with reference to the "Work Sheet for Adjusting Axle Drive".

During the operation "Adjustment of Axle Drive", the axle tube at lock side has already been clamped in trestle, both outer taper roller bearing races have been pressed into axle tubes using special tool No. 20, and inner taper roller bearing races have been forced onto differential housing together with shims.



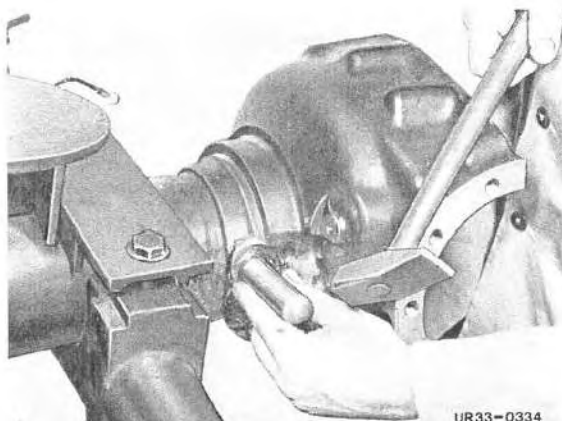
UR 33-0289

- 1 Position axle tubes horizontally in trestle.
- 2 Install selector fork with sliding blocks and shift dog in axle tube.



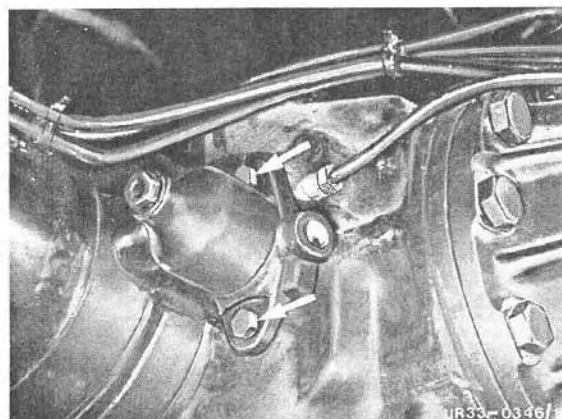
UR 33-0310

747.2



UR33-0334

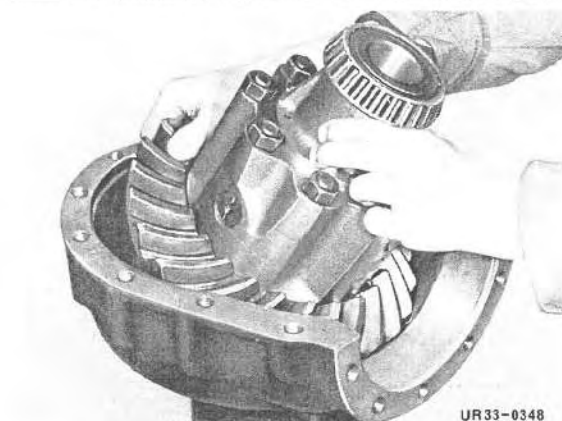
- 3 Knock swivel joint into position, using special tool No. 18.



UR33-0346

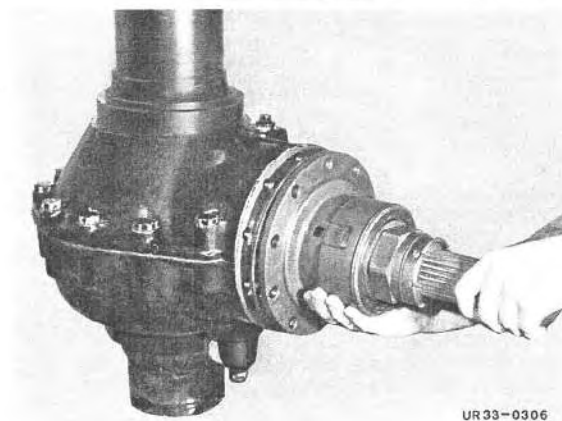
- 4 Attach differential lock shift cylinder.

- 5 Turn axle tubes by 90° in trestle.



UR33-0348

- 6 Install differential in axle tube at lock side.



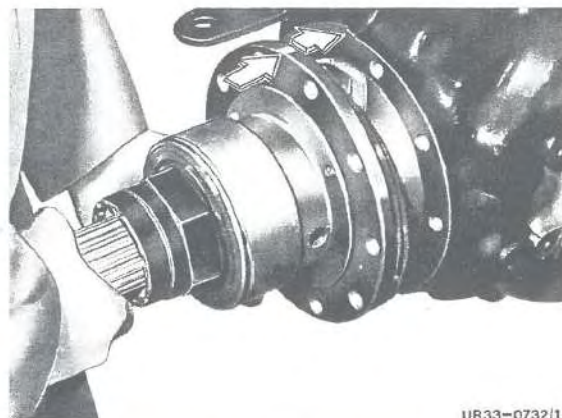
UR33-0306

- 7 Coat both axle tubes with sealant No. 1 and then bolt together.

Note: To center axle housing correctly, insert pinion flange into housing.

8 Coat splines on drive pinion bearing with running - in paste No. 6 and install drive pinion bearing with the shim rings determined on the dimension sheet and sealing ring in axle housing.

Note: Notch and oil duct facing upwards.



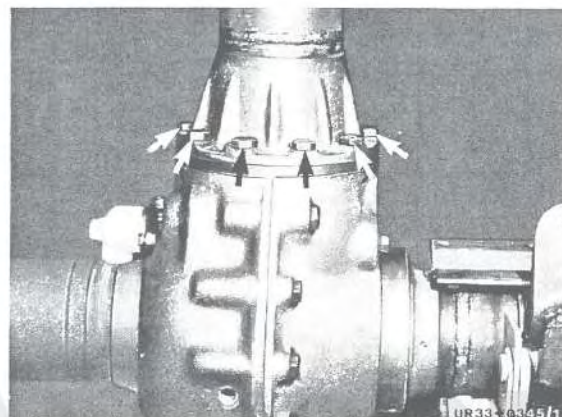
UR33-0732/1

9 Coat torque tube flange with sealing compound No. 1 and bolt torque tube to pinion flange.

10 Turn axle in assembly block 90°.

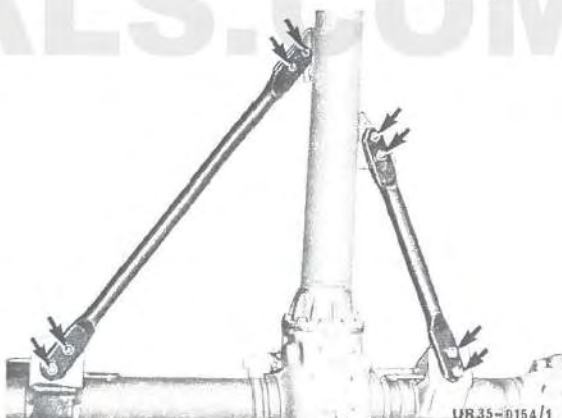
11 Clean splines and cavity of drive shaft, fill cavity behind the profile with lubricating grease No. 2 (50 g.) and install propeller shaft.

12 Assemble wheel hub drive, refer to 6.1/1, coat with sealing compound No. 1 and install.



UR33-0345/1

13 Coat fastening bolts with locking agent No. 4 and attach axle struts.



UR35-0154/1

14 Attach spring brake cylinders, check adjustment a dimension (actuating lever must rest on stop of fixed caliper). Adjust as required by shifting the spring brake cylinder in the oblong holes. Adjustment "a", refer to 1.3/1.

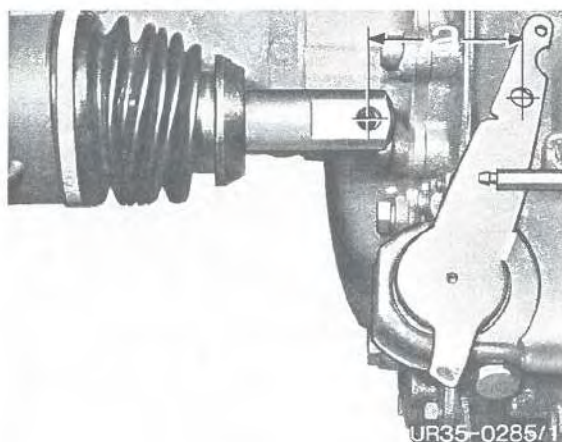
Note: Do not actuate emergency release device.

15 Attach brake and differential lines.

16 Remove axle from assembly fixture.

17 Attach road wheels and tighten.
For tightening torques, refer to 1.5/1.

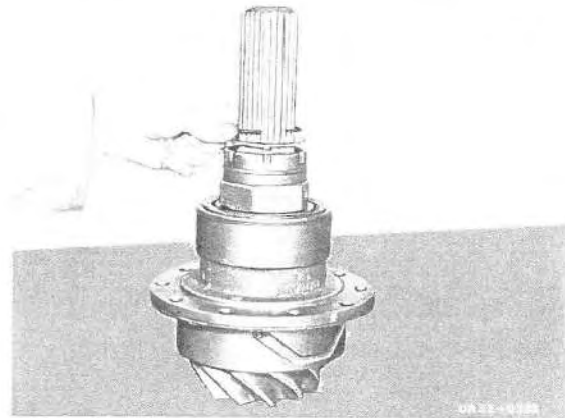
Note: Retighten wheel nuts after 50 km.



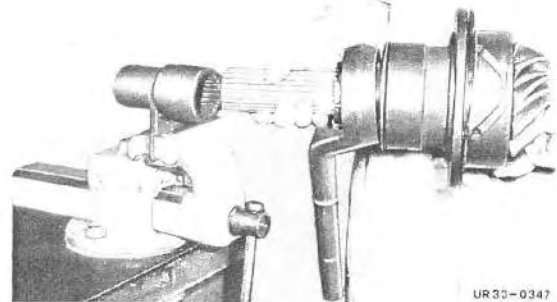
UR35-0285/1

Disassembly

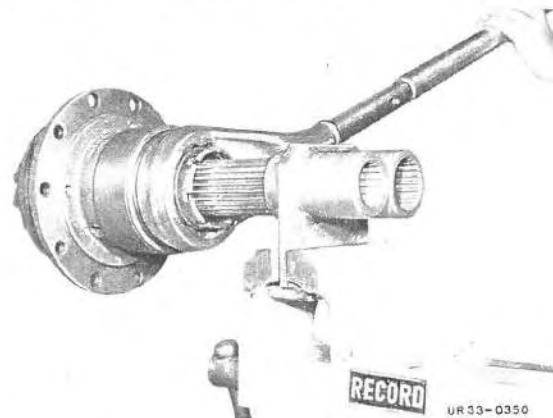
1 Remove lock washer and serrated washer from hexagon nut.



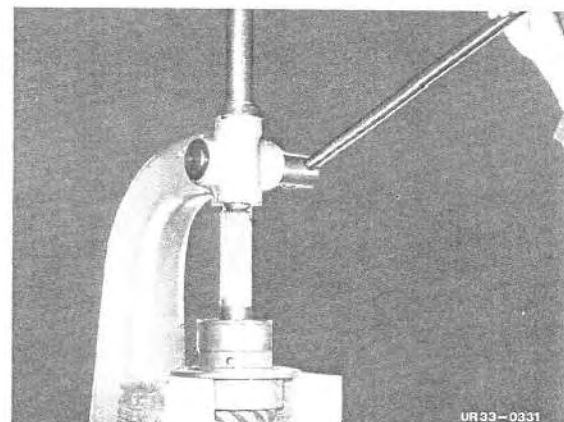
2 Clamp drive pinion in vise using special tool No. 16.

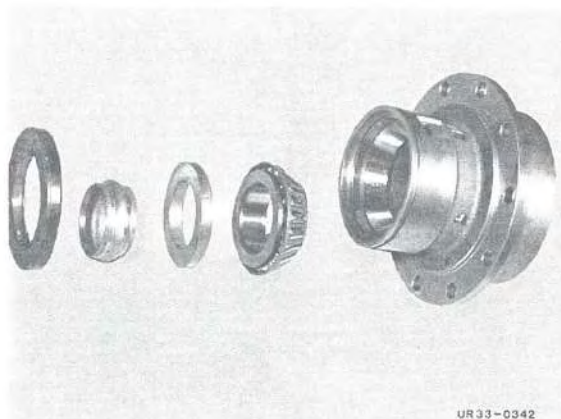


3 Unscrew hexagon nut.



4 Force drive pinion out.





- 5 Remove retaining ring, shaft seal, taper roller bearing and clamping sleeve from pinion flange.



- 6 Draw outer taper roller bearing races out of pinion flange, using special tool No. 9.

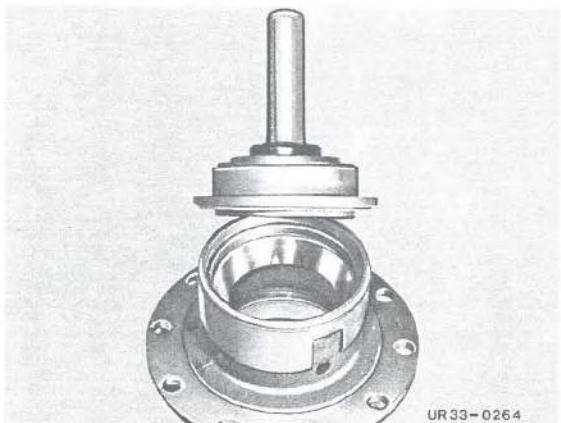


- 7 Draw taper roller bearing and inner race off drive pinion.

- 8 Clean and check all parts, exchange if necessary.

Note: Prior to installing taper roller bearing, rinse out and insert with SAE 90 oil.

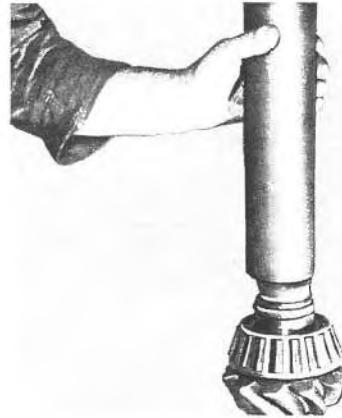
Important: If drive pinion is to be exchanged, it must always be replaced **together** with ring gear. Make sure serial numbers are identical.



Assembly

- 1 Force outer taper roller bearing race into pinion flange, using special tool No. 20.

- 2 Press tapered roller bearing onto drive pinion using special tool No. 17.
- 3 Insert drive pinion into pinion flange.

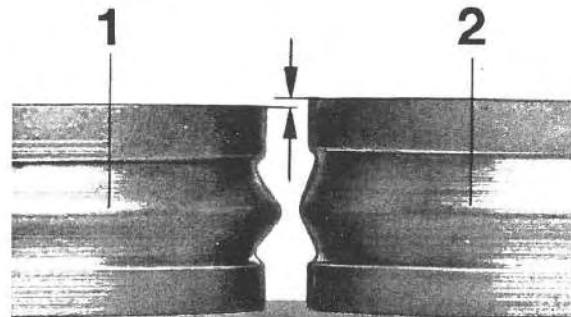


UR 33-0265

- 4 Fit new clamping sleeve on drive pinion.

Note: Use only a new clamping sleeve (2) since the old clamping sleeve (1) is compressed during initial installation.

- 5 Coat tapered roller bearings with lubricating grease No. 2 and press into bearing seat using special tool No. 17 so that bearing has a clearance of 1 to 2 mm.



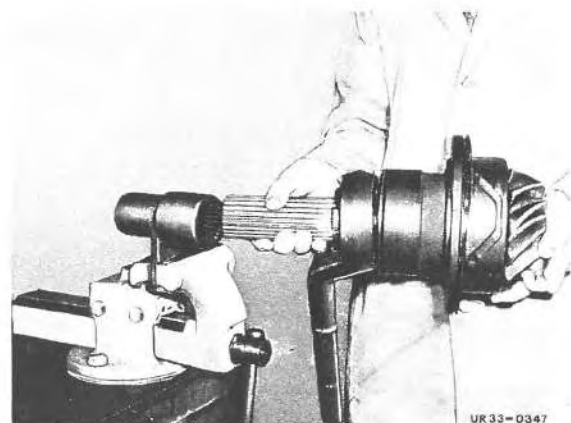
UR 33-0009

- 6 Coat oil seal with lubricating grease No. 2 and install using special tool No. 21.
- 7 Install bearing race, screw on hexagon nut.



UR 33-0289

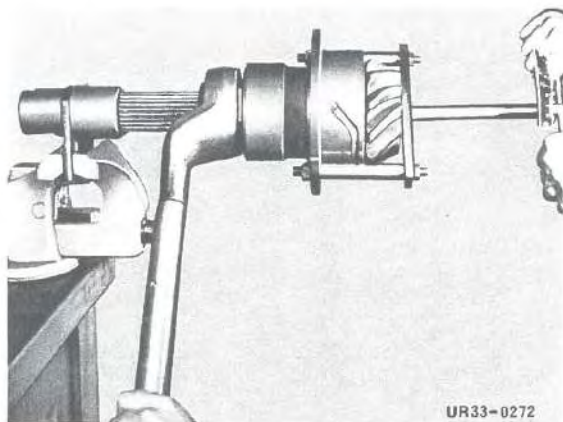
- 8 Clamp drive pinion together with special tool No. 16 in vise and tighten hexagon nut until all the bearing play has been taken up.



UR 33-0347

35.3 Disassembly and Assembly of Drive Pinion Bearing

747.2



Adjusting Drive Pinion

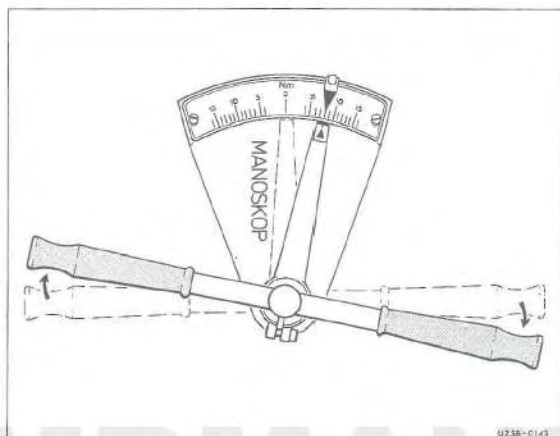
1 Mount special tools No. 14 and No. 15 on pinion flange.

Refer to worksheet 1.7/1.

2 Screw on large nut and tighten until all perceptible play of the flange is eliminated.

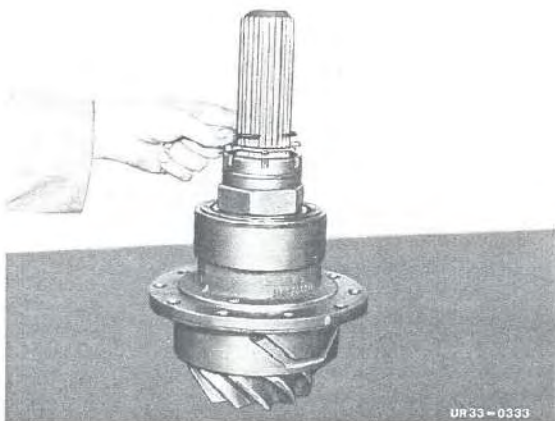
Note: Tighten nut slowly, while constantly checking friction coefficient to ensure the value is not exceeded.

Prior to checking with the torque wrench turn the flange several times by hand.



The preload (breakaway force) is correctly adjusted if the flange begins to turn at a value of 6,0 Nm.

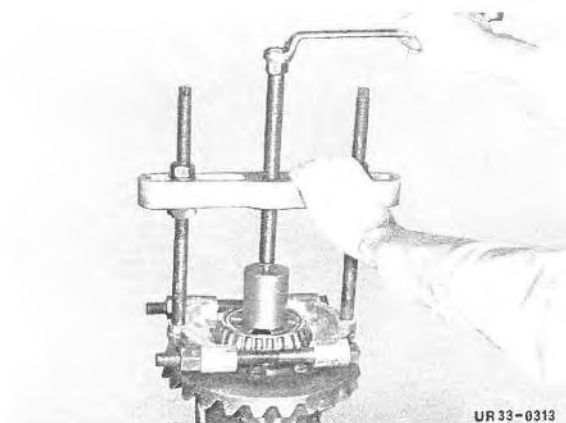
If this value is exceeded, a new clamping sleeve must be installed once again.



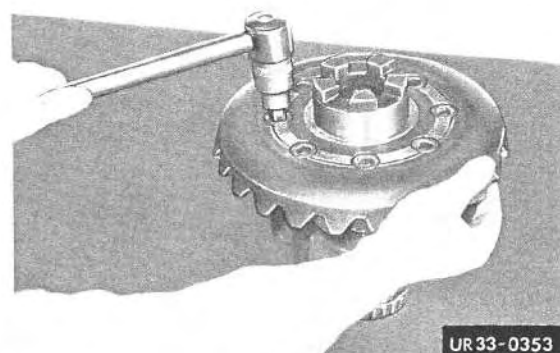
3 Install serrated washer, lock washer and sealing ring in hexagon nut.

Disassembly

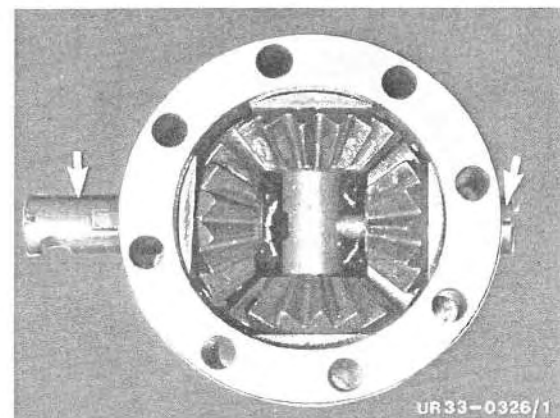
- 1 Remove rear axle, referring to 2.1/1.
- 2 Disassemble rear axle, referring to 3.1/1.
- 3 Using puller, draw both inner taper roller bearing races off differential.



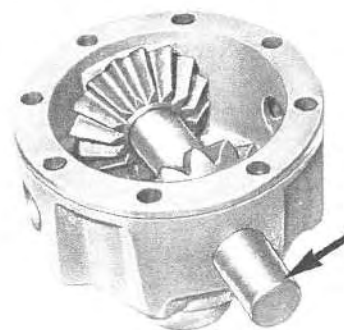
- 4 Unscrew and remove ring gear from differential housing, using special tool No. 3.

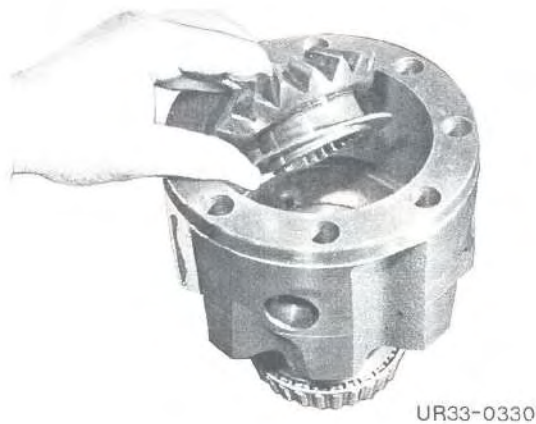


- 5 Remove thrust washer and side gear.
- 6 Remove both short differential pinion shafts.
- 7 Take thrust segments and differential pinions out.



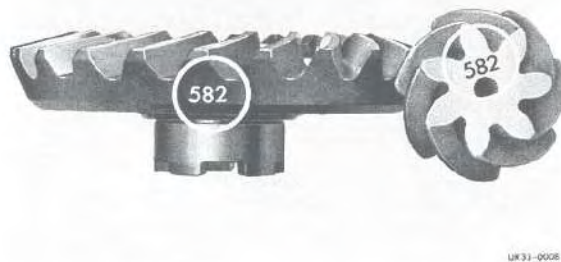
- 8 Remove long differential pinion shaft.
- 9 Take thrust segments and differential pinions out.





10 Take side gear and thrust washer out of differential housing.

11 Clean and check all parts, exchange if necessary.



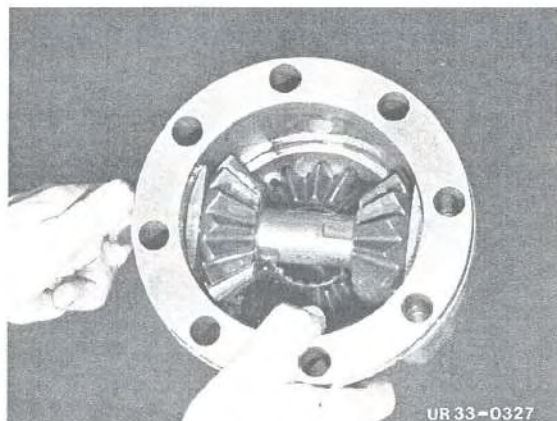
Assembly

Note: If ring gear is to be exchanged, it is always to be replaced together with drive pinion. Make sure serial numbers are identical.

1 Coat all pinion shafts and thrust segments with paste No. 3



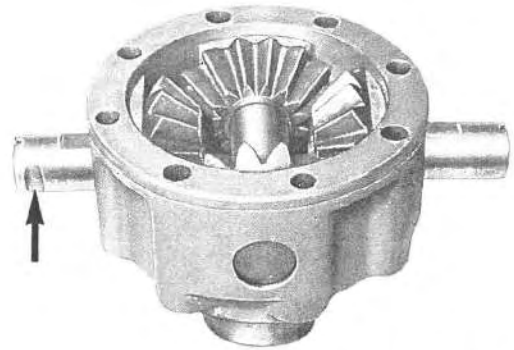
2 Fit thrust washer and side gear in differential housing.



3 Fit thrust segments, differential pinions and differential pinion shafts.

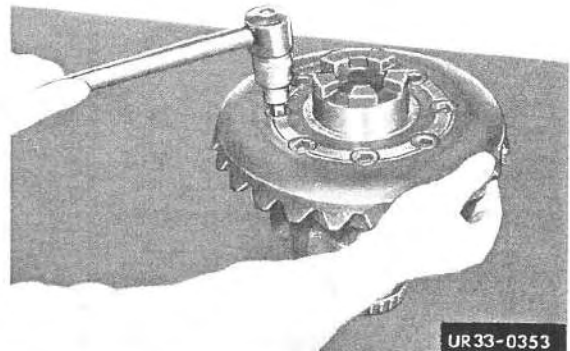
Note: Note recesses in differential pinion shafts for bolts.

4 Place thrust washer and side gear in differential housing.



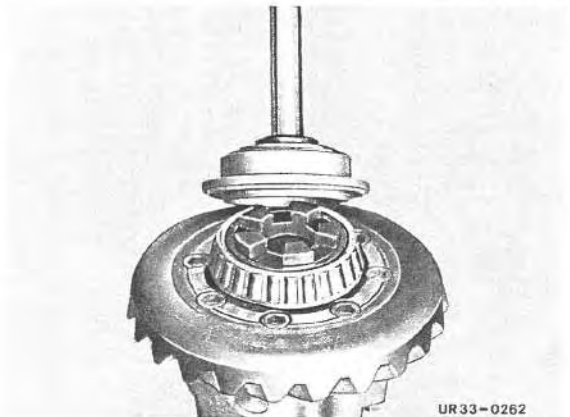
UR33-0473/1

5 Using special tool No. 3, bolt differential housing to ring gear and torque fastening bolts. See 1.5/1 for tightening torques.



UR33-0353

6 Using special tool No. 20, force large inner taper roller bearing race onto differential housing and coat with fluid grease No. 5.

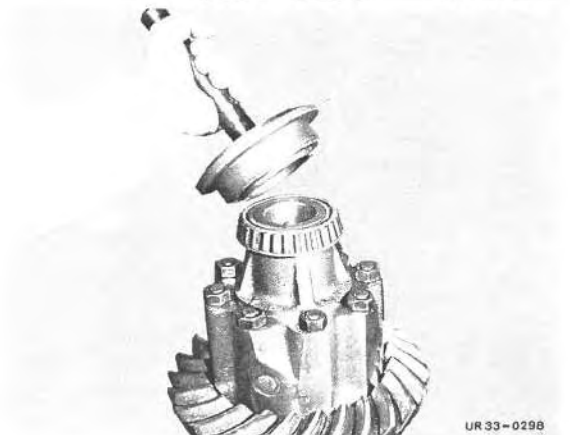


UR33-0262

7 Using special tool No. 20, force small inner taper roller bearing race onto differential housing as far as guide and coat with fluid grease No. 5.

8 Assemble rear axle, referring to 3.1/1.

9 Install rear axle, referring to 2.1/1.



UR33-0298